

Supplementary Analysis: Privacy and Robustness Tests for Vertically Federated IEGHS Scheduling

1 Introduction

This supplementary report gives the experimental settings and detailed results supporting Section 4.4 of the revised manuscript. It includes coordinator-side inference, masked-fusion equivalence, multi-round attacks, security-layer ablation, threshold sensitivity, physical-recovery diagnostics, and software overhead. The concise discussion in the manuscript reports the principal findings, while the material below records the configurations and numerical values used in the additional evaluation.

2 Test System and Experimental Setup

2.1 System and data

All physical calculations in this study originate from `centralized_P118.py`. The coupled system contains an IEEE 118-bus electricity network, a 20-node natural-gas network, and a 7-node hydrogen network. The privacy and robustness study uses 517 feasible scenarios generated from this model. The software implementation is written in Python 3.11 and PyTorch; local physical recovery uses Gurobi through the original model interface.

Three seeds, 2026, 2027, and 2028, are used for the main learning and attack experiments. The dataset is divided once into 60% training, 20% public calibration, and 20% test data. The final test partition contains 104 records. Public calibration data are fixed before private training and are used for attacker fitting, semantic profiles, and post-release attack updates. Final test records are reserved for reported performance.

The [project repository](#) contains this report, machine-readable results, scripts, environment information, and the release commit used for the revision.

2.2 Threat model and protocol

The experiments consider three sources of risk: an active network attacker, an honest-but-curious coordinator, and an authenticated participant that submits altered features or updates. The coordinator follows the training protocol and may collude with one participant. Authenticated keys, uncompromised endpoints, retained counters, and an approved local execution environment are assumed. The reported tests exclude coordinator collusion with two participants, compromised keys, false measurements signed by a trusted source, denial-of-service, and hardware side channels.

AES-256-GCM protects messages in transit. Each entity retains its embedding E_i and first-layer block W_i , computes $z_i = W_i E_i$, and transmits

$$\tilde{z}_i = z_i + \sum_{j>i} r_{ij} - \sum_{j<i} r_{ji}. \quad (1)$$

Pairwise masks cancel in the authenticated aggregate,

$$\sum_i \tilde{z}_i = \sum_i z_i, \quad (2)$$

so the coordinator receives the fused activation required by the upper SplitNN. Local DP-SGD addresses training-record privacy, and representation regularization reduces property information in the fused activation for the evaluated attacker. Authenticated envelopes bind participant identities, session, round, counter, tensor dimension, and payload digest. Trusted local recomputation checks submitted updates. Signed source commitments detect changes made after a trusted source has signed a scenario-round record.

2.3 DP-SGD configuration and accounting

The record-level differential-privacy guarantee applies to the 310-record private-training partition. Per-record gradients are clipped to an ℓ_2 norm of 0.2, followed by Gaussian perturbation with a noise multiplier of 3.0. Poisson sampling uses an expected batch size of 32, corresponding to a sampling rate of $32/310 = 0.1032$.

Private training contains 10 epochs with 10 update steps per epoch, giving 100 DP-SGD steps. Using the Rényi differential-privacy accountant, the resulting privacy budget is $\varepsilon = 1.8892$ at $\delta = 1.0406 \times 10^{-5}$, with optimal Rényi order 13.

Table 1: Record-level DP-SGD configuration for the private-training partition.

Parameter	Value
Protected training records	310
Per-record clipping norm	0.2
Gaussian noise multiplier	3.0
Expected batch size	32
Poisson sampling rate	0.1032
Private-training epochs	10
Total DP-SGD steps	100
Accountant	Rényi DP
Privacy budget ε	1.8892
Failure probability δ	1.0406×10^{-5}
Optimal Rényi order	13

The 103 public-calibration records are excluded from the protected training population. Public pretraining and the representation-defense stage use only these declared public records and their convex synthetic recombinations; they provide no DP guarantee for the public-calibration records and do not incur additional privacy loss for the private-training records. The implementation assumes that per-record gradients remain inside the approved local mechanism. It uses pseudorandom numerical Gaussian noise and does not implement a cryptographically secure distributed-noise-generation protocol.

3 Coordinator-side inference and utility

3.1 Evaluation protocol

The property attacker observes the fused activation available to the coordinator. A logistic-regression classifier and the sensitive-property thresholds are fitted on 103 public calibration records. ROC AUC is then measured on 104 held-out test records. The representation defense is trained with public calibration records and convex recombinations of those records. Final test records are excluded from defense training and configuration selection.

The membership attacker uses the per-record loss as its score. It compares a balanced set of 104 private-training members and 104 held-out test nonmembers. The reported AUC describes this score and these two populations. Embedding inversion, inversion of the returned common fusion derivative, and reconstruction from the released aggregate require separate observation models and are listed in Section 9.

3.2 Per-seed results

Table 2: Coordinator-side property inference, membership inference, and P2H utility.

Seed	Property AUC, base	Property AUC, defended	Membership AUC	P2H RMSE, defended
2026	0.9010	0.5927	0.4754	0.6785%
2027	0.9044	0.5879	0.4784	0.7385%
2028	0.8629	0.6316	0.4933	0.9722%
Mean	0.8894	0.6041	0.4824	0.7964%

The mean property-inference AUC falls by 0.2853, from 0.8894 to 0.6041. The defended values remain above random guessing and show residual property information in the fused representation. The mean P2H relative RMSE changes from 0.8836% to 0.7964%. The loss-based membership attacker gives a mean AUC of 0.4824.

3.3 Numerical equivalence of masked fusion

Table 3: Maximum absolute differences between plaintext and masked fusion.

Quantity	Maximum absolute difference
Prediction	1.91×10^{-6}
Loss	3.73×10^{-9}
Gradient	1.49×10^{-8}
SGD update	7.45×10^{-9}

These differences are at floating-point scale. The privacy-utility change reported in Table 2 is associated with representation regularization and private training, while masked fusion preserves the intended first-layer sum.

4 Multi-round active attacks

4.1 Attack construction

Each case runs for 20 rounds with batch size 64 and learning rate 0.005. Every seed is combined with the power, gas, and hydrogen entities and six attack types, producing 54 cases. Post-release adaptive updates use only the public calibration partition.

Table 4: Active attacks used in the 54-case experiment.

Attack	Construction
Feature poisoning	The first eight local features are reflected and shifted by 4.0. The defended path clips each feature to the public 1st–99th percentile interval.
Gradient sign flip	The submitted update is -8 times the clean update.
Model replacement	The submitted update is 30 times the clean update.
Gaussian Byzantine	A random Gaussian direction is scaled to 15 times the clean-update norm.
Trigger backdoor	Twenty local optimization steps with learning rate 0.005 construct the submitted backdoor update.
Threshold-aware slow drift	An orthogonal random direction is added with scale 2.5 relative to the clean-update norm.

Trusted local recomputation discards the submitted update for the five update-space attacks and recomputes the prescribed update inside the approved local execution environment. Feature poisoning enters before this stage, so its defended path uses public-range clipping.

4.2 Aggregate results

The 54 cases finish without numerical divergence. Trusted recomputation reproduces the clean trajectory for each tested update-space attack. Feature poisoning retains a mean RMSE of 1.2919% after range clipping, which motivates the source-integrity and semantic tests in the next section.

Table 5: Mean P2H relative RMSE over nine cases per attack.

Attack	Clean	Attacked	Defended
Feature poisoning	0.7449%	2.1193%	1.2919%
Gradient sign flip	0.7449%	9.6109%	0.7449%
Model replacement	0.7449%	1.5209%	0.7449%
Gaussian Byzantine	0.7449%	0.7459%	0.7449%
Trigger backdoor	0.7449%	3.4045%	0.7449%
Threshold-aware slow drift	0.7449%	0.7448%	0.7449%

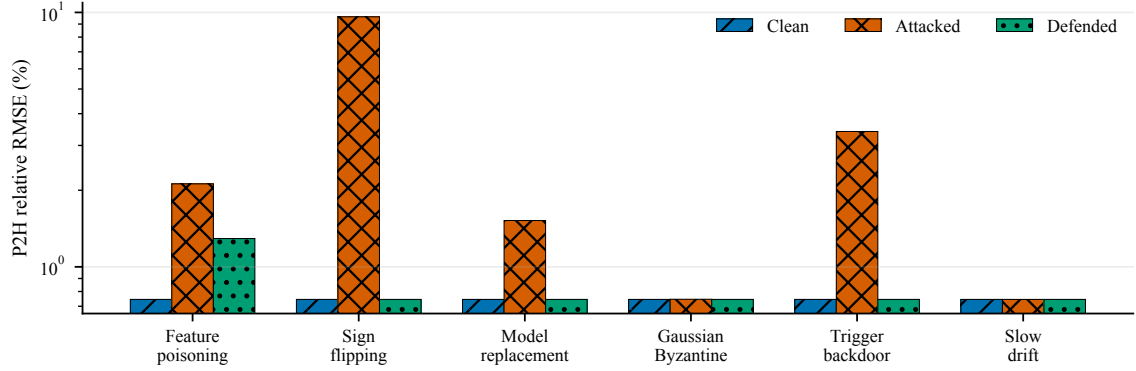


Figure 1: P2H prediction accuracy under six 20-round active attacks across three entities and three seeds.

5 Semantic feature attacks and layer ablation

Four feature-space attacks are applied within public feature ranges: a six-hour temporal roll, an in-range drift ramp covering 30% of the public range, cross-scenario replay, and eight-step white-box PGD. PGD uses an L_∞ radius equal to 25% of the public feature range and clips each iterate to both the attack region and public bounds. The experiment combines three seeds, three entities, and four attacks, giving 36 attack cases. Each case runs for 20 rounds.

The ablation introduces six layers in sequence: secure aggregation, authenticated transport, trusted local recomputation, range clipping, temporal/manifold screening, and an Ed25519 source commitment. The semantic profile uses the public 1st–99th percentile feature and temporal-difference ranges and the 95th percentile of public nearest-neighbour distances.

Table 6: Layer-wise semantic-attack results.

Layer	Mean P2H RMSE	Parameter difference	Rejection rate
Secure aggregation	1.2461%	1.811×10^{-3}	0
Authenticated transport	1.2461%	1.811×10^{-3}	0
Trusted recomputation	1.2461%	1.811×10^{-3}	0
Range clipping	1.2461%	1.811×10^{-3}	0
Temporal/manifold screening	1.1958%	1.078×10^{-3}	0
Signed source commitment	0.7449%	0	100%

Authenticated communication and trusted recomputation preserve the computation associated with the received feature values. Semantically altered inputs can remain admissible under numerical range checks. Temporal/manifold screening lowers the average error, with attack-dependent performance. Source commitment recovers the clean trajectory for every tested modification made after commitment. This result relies on a trustworthy signing source.

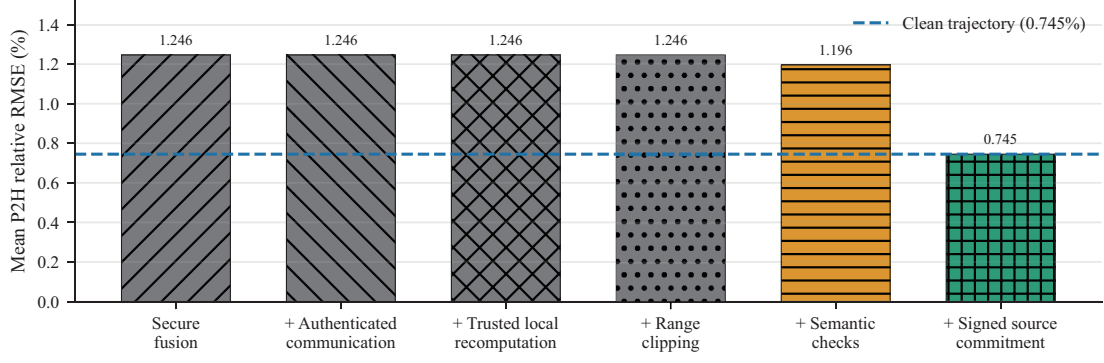


Figure 2: Layer-wise security ablation under four semantic feature attacks.

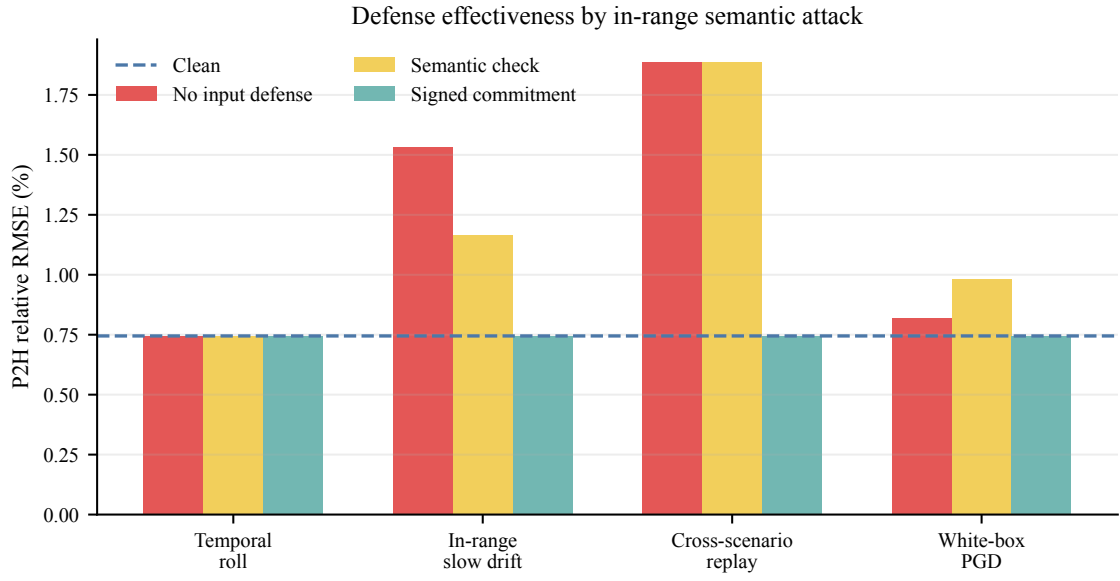


Figure 3: Attack-specific performance of semantic screening and signed source commitment.

6 Sensitivity of semantic screening

The fixed grid combines two-sided tail probabilities of 0.5%, 1%, and 2.5% with public-manifold quantiles of 0.90, 0.95, and 0.99. The test uses seed 2026, the hydrogen participant, 20 rounds, and the four semantic attacks. The nine settings are declared before final-test evaluation.

Attack RMSE without screening is 2.1977%. Projection gives 2.0061%–2.0114%, while the same operation changes clean-data RMSE from 0.7449% to 0.7564%–0.7580%. All tested thresholds modify some clean inputs, and none restores the clean attack trajectory. The operational role of this stage is anomaly screening; source commitment supplies the integrity decision under the trusted-source assumption.

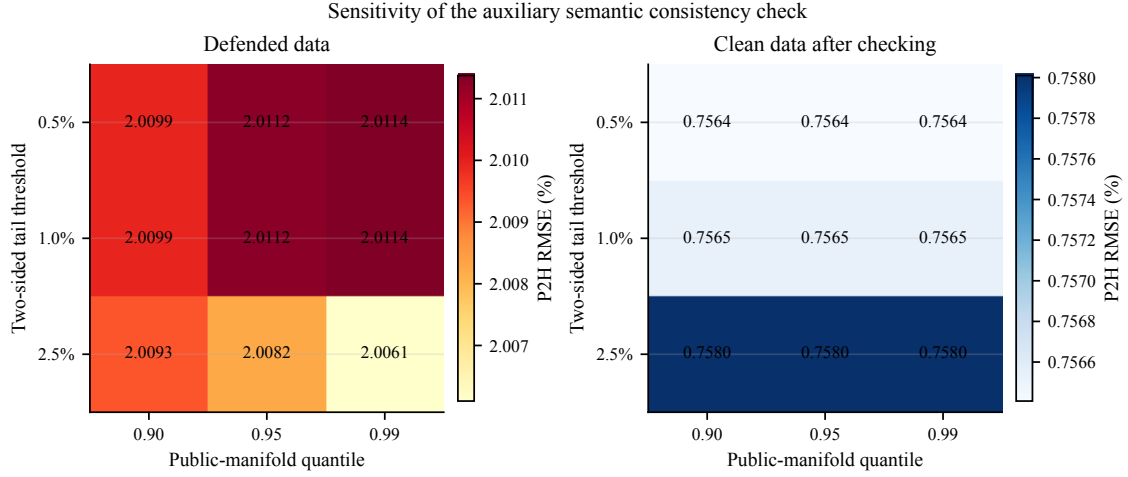


Figure 4: Sensitivity of semantic screening to tail and public-manifold thresholds.

7 End-to-end physical recovery

The physical-recovery test fixes 30 scenarios before evaluating final metrics. The selected stress case uses seed 2026, the hydrogen participant, and white-box PGD. Clean, attacked, and semantically screened coupling predictions are passed to the original local recovery procedure. The penalty parameter is 3×10^6 , the maximum iteration count is 20, and the consensus and numerical-quality limits remain 0.01.

Table 7: Fixed 30-scenario physical-recovery diagnostics.

Metric	Clean	PGD attack	Semantic screening
Completed recoveries	30/30	30/30	30/30
Consensus convergence	30/30	30/30	30/30
Quality-limit pass	30/30	30/30	30/30
All-party Gurobi status 2	30/30	30/30	30/30
Maximum constraint violation	3.40×10^{-7}	8.79×10^{-7}	7.32×10^{-7}
Mean final primal residual	0.00653	0.00686	0.00569
Mean schedule deviation from clean	—	0.1528%	0.1834%
Maximum absolute cost deviation	—	0.2017%	0.2240%
Mean recovery time	27.97 s	28.17 s	28.18 s

Every group satisfies the original physical limits. The PGD and screened schedules retain small deviations from clean dispatch. Semantic projection slightly increases the mean schedule deviation in this test. Physical recovery secures feasibility, while the reported dispatch and cost changes retain part of the attack’s operational effect.

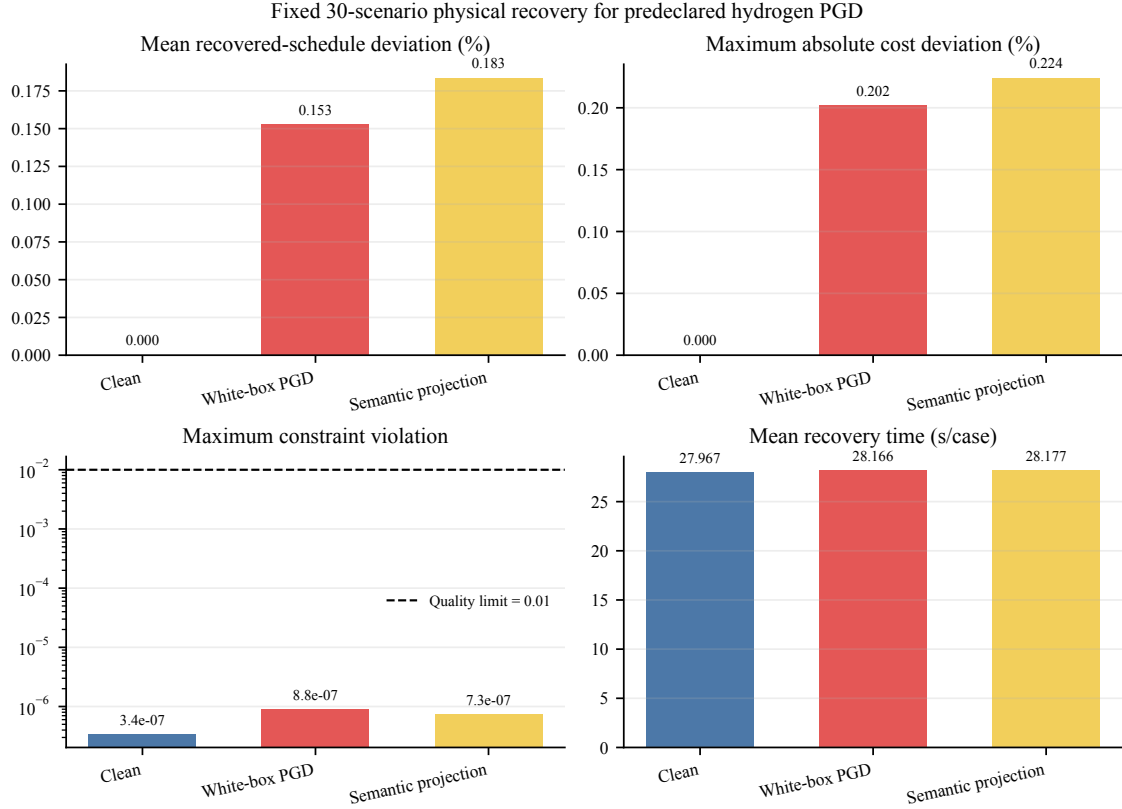


Figure 5: Recovery quality, schedule deviation, and runtime for the fixed 30-scenario test.

8 Software protocol overhead

Latency is measured for the software implementation on the study platform. Communication volume is calculated from the transmitted tensors and authenticated metadata. Source-commitment values are measured separately and added to the authenticated secure-fusion path. The timing figures exclude hardware TEE attestation, wide-area communication, and hardware cryptographic devices.

Table 8: Per-round software latency and communication volume.

Configuration	Latency	Communication
Secure fusion	2.484 ms	4,914 bytes
Authenticated envelopes and recomputation binding	2.932 ms	6,534 bytes
Source commitment increment	0.451 ms	108.9 bytes
Complete software path	3.383 ms	approximately 6,643 bytes

Authenticated envelopes and computation binding add 0.448 ms and 1,620 bytes to secure fusion, corresponding to 18.0% latency growth and 32.96% communication growth. The complete software delay is four orders of magnitude below the approximately 28-s mean physical-recovery time.

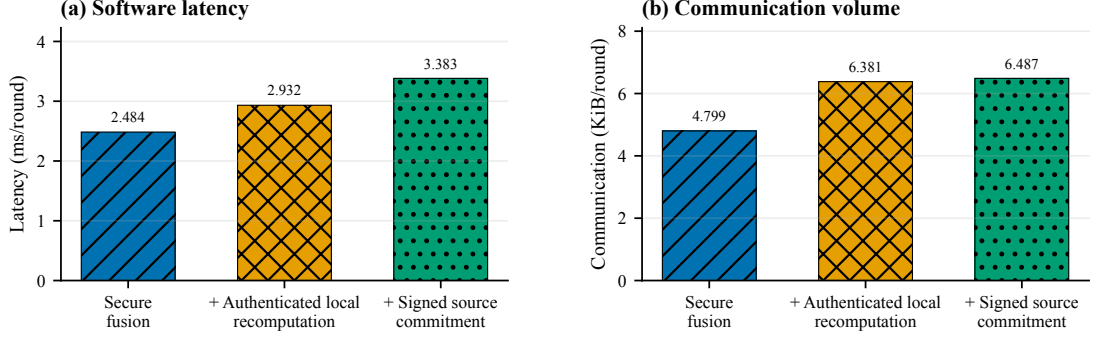


Figure 6: Software latency and communication volume of the layered protocol.

9 Summary and Evaluation Scope

The experiments support four findings. First, pairwise masking preserves numerical output while individual first-layer contributions remain local. Second, representation regularization lowers the property-inference AUC for the tested coordinator-side classifier and maintains P2H prediction accuracy. Third, trusted local recomputation handles the tested update-space attacks, while source-feature corruption requires controls placed before local update generation. Fourth, physical recovery maintains the original feasibility criteria under the selected attack, although small dispatch and cost changes remain.

The inference study covers logistic-regression property inference and loss-based membership inference. Future tests will consider embedding inversion from the fused activation, inversion of the common fusion derivative, reconstruction from the released aggregate, stronger adaptive property attackers, and coordinator collusion with two entities. Source commitment assumes an uncompromised signing source. The software overhead results describe the current implementation and exclude hardware deployment costs. Machine-readable result files and figure-generation scripts are provided in the project repository.